Transformer Optimization Closes Most of the Gap

Abstract

This report revisits an earlier Chinese name generation result. The original matched-pair experiment appeared to show a large LSTM advantage over a small causal Transformer. Reviewer feedback identified a serious optimization confounder: the Transformer had been trained with flat Adam and no warmup schedule.

After adding Transformer-tuned baselines with AdamW, 10% linear warmup, cosine decay, and lower dropout, the Transformer test perplexity improved from 86.71 to about 79.03. The LSTM still has a small residual edge at 77.92, but the original large gap is mostly explained by optimization rather than architecture alone.

Updated thesis

The defensible conclusion is no longer "LSTM beats Transformer". It is: on this dataset and model scale, proper Transformer optimization closes most of the original gap; under the tested tuned recipes, the LSTM retains a small, stable perplexity edge.

New Main Result

This figure replaces the older headline chart. It shows why the previous conclusion was too strong: the apparent 8.79 PPL architecture gap collapses to roughly 1.1 PPL once the Transformer receives a more appropriate optimization recipe.

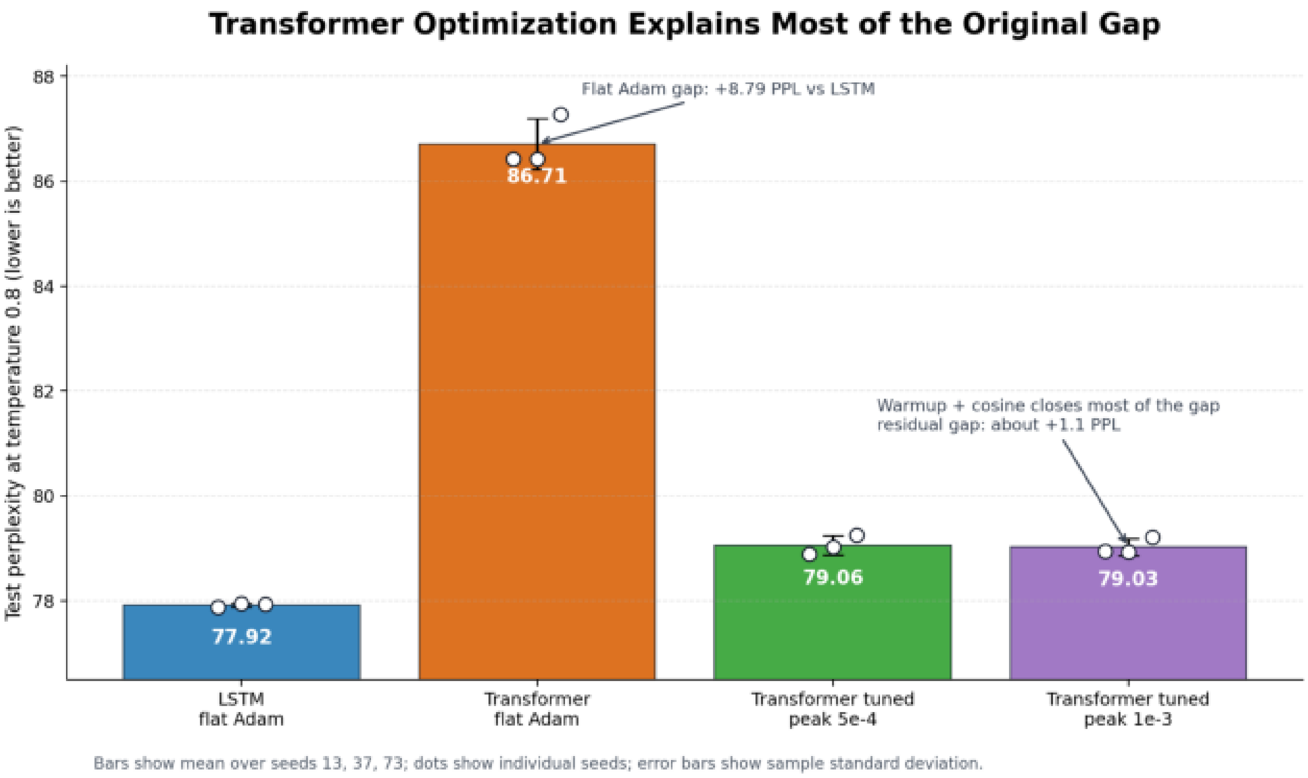

Figure 1. Full-data test perplexity at temperature 0.8. The original flat-Adam Transformer is a weak baseline; warmup plus cosine decay closes most of the gap. The LSTM still remains slightly lower in this run set.

Full-Data Results After Tuning

All values use the full training split, seeds 13/37/73, and generation temperature 0.8.
Perplexity is exp(mean cross-entropy over non-PAD target tokens), so lower is better.

Model / recipe	Test PPL mean +/- std	Distinct-2	Novelty	Train overlap
LSTM flat Adam	77.9209 +/- 0.0342	0.8851	0.4900	0.5100
Transformer flat Adam	86.7124 +/- 0.4884	0.8383	0.5095	0.4905
Transformer tuned peak	79.0579 +/- 0.1839	0.8655	0.4688	0.5312
Transformer tuned peak	79.0286 +/- 0.1572	0.8625	0.4773	0.5227

The two tuned Transformer runs use the same architecture as transformer_small. Only the optimization recipe changes: AdamW, weight decay 0.01, 10% warmup, cosine decay, dropout 0.1, and peak LR 5e-4 or 1e-3.

Experimental Controls and Configurations

Data and vocabulary

The source is the China Biographical Database. The cleaned study dataset contains 584,442 full names with an 80/10/10 train/validation/test split. The split seed is 20260419; training seeds are 13, 37, and 73.

Tokenization is a naive character-level mapping over single Unicode characters plus SOS, EOS, PAD, and UNK. No BPE, unigram tokenization, or subword regularization is used. The full-data shared vocabulary is about 8.4k tokens.

Architecture

The LSTM-small-matched model has embedding dimension 160, hidden size 320, two strictly forward LSTM layers, dropout 0.4, and 5,469,500 trainable parameters.

The Transformer-small model has embedding dimension 256, two causal self-attention layers, eight heads, feed-forward dimension 512, learned absolute positional embeddings with max_seq_len 10, and 5,355,708 trainable parameters.

Optimization recipes

The original baseline used flat Adam with learning rate 0.001 and dropout 0.4 for both neural models. This is now treated as a training-recipe baseline, not as a fair best-effort Transformer baseline.

The tuned Transformer baselines use AdamW, weight decay 0.01, dropout 0.1, 10% linear warmup, cosine decay, and peak learning rates 5e-4 or 1e-3. Both tuned runs keep the architecture and parameter count fixed.

Original Convergence Context

The original Transformer plateau was real under the flat-Adam recipe, but the tuned baselines show that the plateau was largely an optimization artifact.

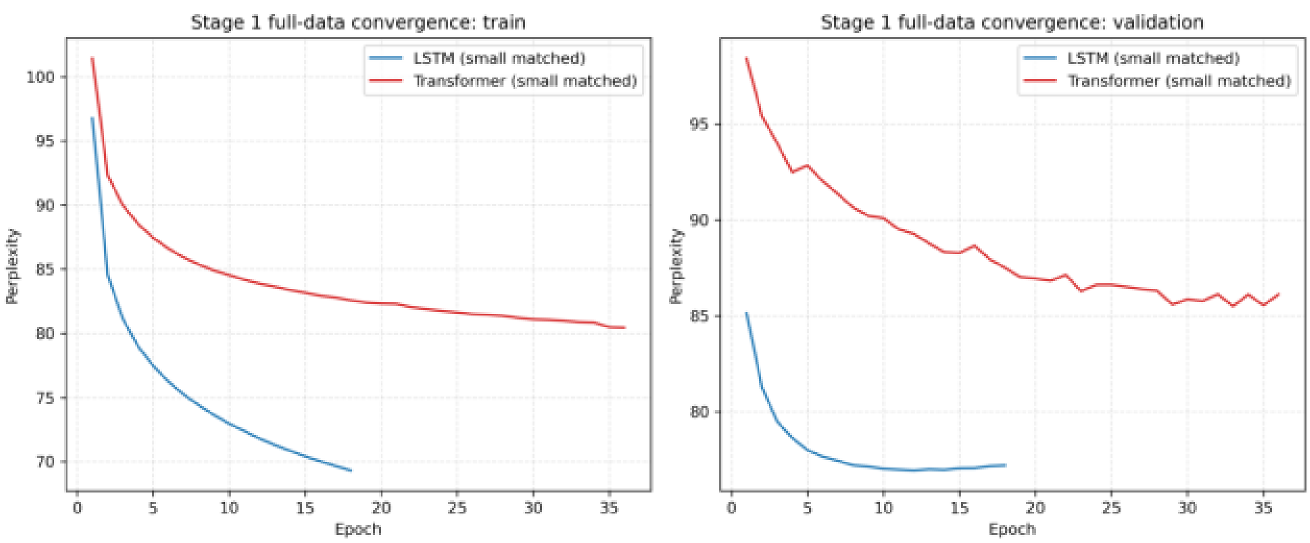

Figure 2. Original flat-Adam convergence curves. These curves are still useful historically, but they should no longer be used alone to argue for an architectural LSTM advantage.

Original Data-Scale Ablation

The next necessary extension is to repeat the data-scale ablation with the tuned Transformer recipe if the project wants a stronger architecture-level claim.

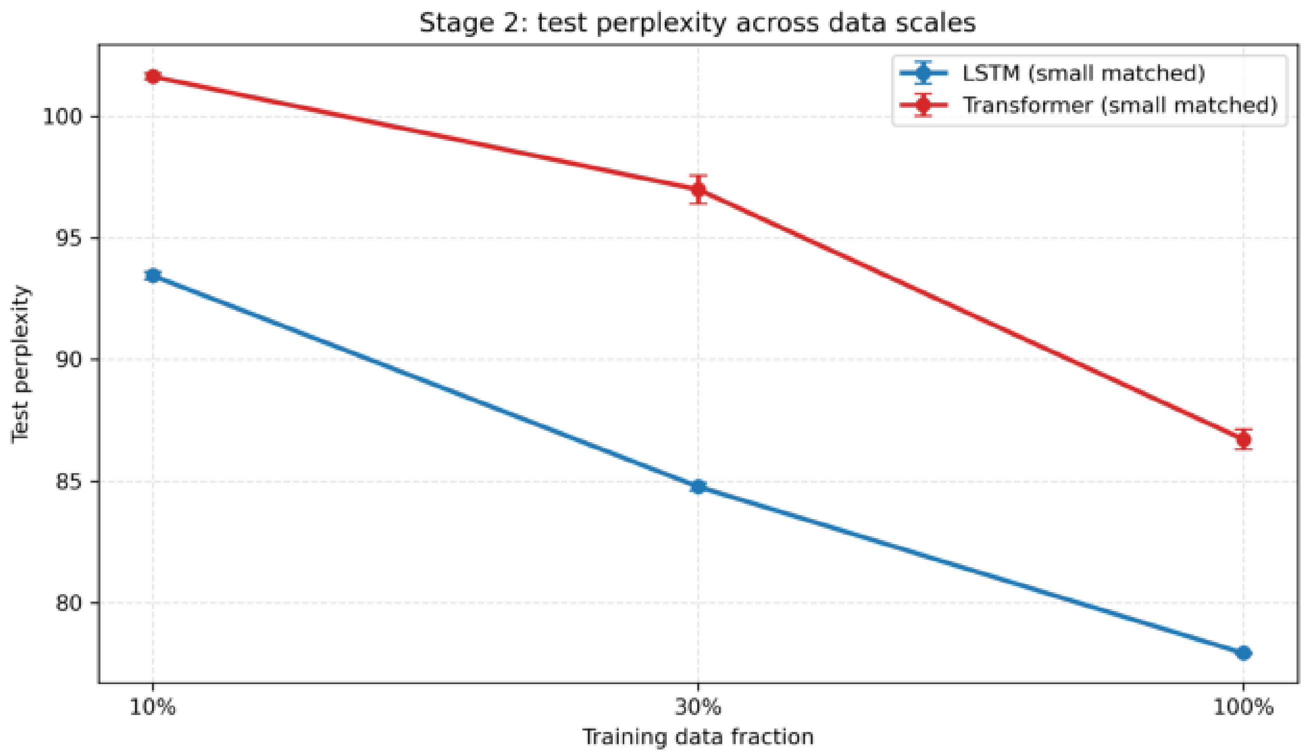

Figure 3. The previous data-scale ablation still describes the flat-Adam comparison. It should be interpreted as evidence about the original shared recipe, not as a final architecture verdict.

Metric and Decoding Definitions

Teacher-forced metrics

Cross-entropy is averaged over non-PAD target tokens. Perplexity is exp(cross_entropy), using natural-log base e.

Generation metrics

unique_name_ratio is unique generated full names divided by total generated names. Distinct-1 is unique generated characters divided by total generated characters. Distinct-2 is unique generated character bigrams divided by total generated character bigrams.

novelty_ratio is the share of generated names not appearing exactly in the training set. train_overlap_ratio is the exact-overlap complement. duplicate and invalid-output ratios track local validity failures.

Decoding

Reported generation metrics use pure temperature-scaled categorical sampling at temperature 0.8. No top-k truncation and no nucleus/top-p sampling are used in these reported metrics.

Revised Interpretation and Next Steps

Interpretation

The original large gap should not be presented as an architecture result. It is mostly an optimization result: a flat-Adam Transformer baseline was substantially weaker than a tuned Transformer baseline.

The residual result is still interesting. Across the tested tuned recipes, the LSTM remains about 1.1 perplexity points lower than the Transformer. That is much smaller than before, but it is consistent across seeds in these runs.

What remains open

A stronger study should rerun the 10% and 30% data-scale ablations with the tuned Transformer, test additional dropout values, and possibly test weight tying because the input/output embedding and projection layers consume a large fraction of the parameter budget.

The public claim should be cautious: tuned optimization closes most of the Transformer gap; a small LSTM edge remains in this setup. Anything stronger would require the next ablation round.